Finding the Best Keyboard Layout

Search, Bounds and Certificates

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Working draft skeleton · no results yet · bracketed TODOs are placeholders
Focus paper 3. Topics T2, T10. Status: eight optimizers and bounds exist; exact windows (#43) and the final "best layout" runs do not.

Abstract

We ask what the best layout is under a fixed cost function, and how sure we can be. We compare eight optimizers (hill climbing, SA, breakout local search, tabu, GA, memetic, MAP-Elites, NSGA-II) under equal budgets on a quadratic assignment problem with sparse trigram terms, give a Gilmore–Lawler lower bound with dual ascent (90.3 % gap at 30 free keys, 1.7 % on the bigram part at 10 free keys), and certify solutions window by window with exact search [TODO]. We report the best layouts found for [TODO: ANSI-30, Corne-36, with layers] with their per-metric breakdowns, compare them with Graphite, Sturdy, Semimak and Canary, and state how far from optimal they can be shown to be.

1. Research questions

1. RQ1 Which optimizer reaches the best layout fastest, by budget (anytime curves, performance profiles, time-to-target over 30 seeds)?
2. RQ2 How large is the provable gap; how does it shrink with fewer free keys?
3. RQ3 Are heuristic optima window-optimal (every k-key subset, k = 6..12, solved exactly)?
4. RQ4 What is the best layout per board/corpus/profile, and how does it differ from the references on each metric?
5. RQ5 Does the answer change with symbols, layers and magic keys (code corpus, 36-key board)?

2. Hypotheses

H1 BLS and SA tie at long budgets, BLS leads at short ones (matches the 2 s leaderboard). H2 $\geq$ 95 % of windows with k $\leq$ 10 are exactly optimal for the best heuristics. H3 misses are trigram-dominated. H4 the global best equals the multi-seed consensus best. All can fail.

3. Method

- Algorithms under the `Optimizer` trait; `gecko bench --seeds 30` over a budget grid.
- Bound: QAP form (`Scorer::qap_terms`), trigrams relaxed to pairs, GL plus 20 dual-ascent iterations; pins as constants.
- Exact: branch and bound; MIP/CP-SAT oracle (#43) for larger windows.
- Reproducibility: every cited layout comes from a run verified with `gecko verify`.

4. Existing evidence

2 s leaderboard (10 seeds): BLS 17.616 $\pm$ 0.081, SA 17.787 $\pm$ 0.145, memetic 18.027 $\pm$ 0.352, tabu 18.960 $\pm$ 0.256 (`bench/memetic-2026-10-21/`); SA 20 s optimum 15.704 under the current profile with seeds agreeing; bound table and ascent convergence 4.49 $\rightarrow$ 5.81 (`context/algorithms/bounds.md`); neighbourhood study (`context/algorithms/README.md`).

5. Figures

F1 anytime curves with CIs · F2 performance profile · F3 gap vs free keys · F4 window-optimality share by algorithm and k · F5 best layout keyboard + per-metric waterfall vs Graphite · F6 best layouts across boards and corpora.

6. Threats

"Best" is relative to the cost function (see paper 2); weak bound at full size; machine-dependent timings; one language unless the multilingual runs are added.

7. Needed

#43 oracle; window driver; 30-seed budget grid; final runs saved as curated `bench/` directories.